Supplementary Material for From Cultivation to Care: Edge IoT, Digital Supervision and Mobile Interfaces for Medicinal-Plant Traceability

---

Version and interpretation note

This supplement retains detailed evaluations and the earlier PLANT2CLINIC clinical prototype as supporting material. The currently linked Health2Clinic demonstrator is a separate application: the earlier prototype's clinical agents and emergency-unblinding features must not be inferred to exist in Health2Clinic. Current mobile interfaces and their inspection scope are documented in the main article and Supplementary Demonstration Guide. All clinical examples use simulated or fictitious data.

The S2 validation script audits values and thresholds in logged anomaly objects, reconstructs severity from recorded anomaly lists, and reuses combined-condition labels. Its zero discrepancies establish record consistency, not independent completeness of anomaly detection. The checked denominators are 136,005 recorded anomaly-band decisions and 54,982 aggregated states; 284 OFFLINE records are excluded from the latter. The 0.24 ms median comes from a separate 100-case probe.

1. Clinical Trial Module: Prototype Extension

The platform has been extended beyond the agronomic module with a functional prototype of a double-blind randomised controlled trial (RCT) clinical module. This extension implements the downstream stages of the PLANT2CLINIC pathway using simulated data: starting from the validated concentrate received from the agronomic module dossier, the clinical module

covers pharmaceutical preparation, confidential randomisation, blinded dispensation, clinical outcome recording, and controlled unblinding. No real patient data are collected or analysed; all clinical records in the current prototype use pseudonymised identifiers (PAC-xxxx codes) and simulated outcome values.

The clinical pipeline is organised into five sequential phases:

1. **Phase 1 — Pharmaceutical Preparation**: The hospital pharmacist prepares extract and placebo kits (KIT-xxxx) by diluting the validated concentrate. Kit type and cannabinoid content are stored encrypted.

2. **Phase 2 — Confidential Randomisation**: Each patient pseudonym (PAC-xxxx) is assigned to treatment arm A (extract) or B (placebo) using a balance-minimisation algorithm.

3. **Phase 3 — Blinded Dispensation**: The medical investigator dispenses a kit to the patient without knowledge of its type.

4. **Phase 4 — Clinical Outcome Recording**: At each visit, the investigator registers patient-reported outcome measures and adverse events.

5. **Phase 5 — Controlled Unblinding**: Emergency unblinding requires dual authorisation. Final unblinding at trial closure requires dual QA authorisation.

The double-blind access model is enforced through a strict role-based domain separation. The *Confidential Zone* (pharmacist role) holds encrypted kit types and arm assignments. The *Blind Zone* (investigator role) accesses only neutral KIT codes, dispensation events, and outcome records.

2. Clinical Agents: C-Trace, BIM, and S2-C

The clinical trial module has been further extended with three deterministic agents that apply the S1–S2–S3 functional paradigm to the clinical domain.

C-Trace (End-to-End Traceability Reconstructor) reconstructs the complete traceability chain from a clinical outcome event back to the agronomic

origin. Given a patient identifier, C-Trace traverses the dispensation record, kit assignment, pharmaceutical concentrate, extraction batch, harvest lot, cultivation room, and genetic variety.

BIM (Blinding Integrity Monitor) analyses the distribution of adverse events and clinical outcome measurements across treatment arms to detect statistical patterns that may constitute a functional unblinding risk. BIM applies two complementary tests: a Fisher exact test on adverse-event contingency tables, and Welch's t-test on per-visit outcome distributions.

S2-C (Clinical Outcome Monitor) applies deterministic threshold-classification logic to longitudinal patient-reported outcomes. For each patient visit, S2-C evaluates the reported scores against published validated-scale thresholds and classifies each patient as `OK`, `WARNING`, `ALARM`, or `ADHERENCE_RISK`.

3. Project Web Page and Resources

A live demonstration of the platform is accessible at `https://trazabilidad.tecnologiashabilitadoras.cloud/`. Note that access credentials are required to use the platform; these can be requested through the corresponding author.

4. Detailed Agronomic-Module Tables

This section collects detailed tables supporting the agronomic-module evaluation reported in the main article. They are provided here to keep the main text focused while preserving full auditability.

4.1. Retrieval-Metric Characterization of the S1 Vector Index

The main-text evaluation reports the correctness of the *answer* returned by the S1 pipeline. Here we complement it with standard information-retrieval metrics computed directly on the vector retriever (`searchRag`), treating it as an isolated component. For each of the 100 evaluation questions we retrieved the

Table S1: Autonomy and safety boundaries of the proposed S1–S2–S3 agentic architecture.

Layer	**Agentic function**	**Autonomy level**	**AI nature**	**Modifies records?**	**Human gate**
S1 Knowledge and Evidence Agent	Evidence retrieval, structuring, and grounded explanation	Advisory	Local RAG + Llama3:8B	No	Review recommended
S2 Operational Monitoring Agent	Telemetry validation and operational-state classification	Deterministic classification	Policy-based, non-generative	No	Operator/QA review
S3 Audit, QA, and Handoff Layer	Validation, accountability, and traceability registration	Procedural validation	Human-in-the-loop workflow	Yes, after validation	Mandatory

Table S2: Catalogue of services implemented in the proposed local-first digital platform prototype.

Service	**Main inputs**	**Main outputs**	**Role in the prototype**
Traceability Passport	QR/batch ID, lineage records, local mirror	Batch passport, lineage, events, documents	Source of operational truth
IoT Monitoring	MQTT telemetry, room metadata, thresholds	Room snapshots, status, alerts	Environmental and fertigation context
S1 Knowledge and Evidence Agent	User query, SOPs, validated documents, QR/batch context, IoT snapshots	Structured evidence block, grounded answer, sources, uncertainty statement	Local analytical retrieval and explanation
S2 Operational Monitoring Agent	IoT snapshots, policies, historical readings, freshness rules	OK/ WARNING/ ALARM/ STALE/ OFFLINE classification, severity, alarm reason	Deterministic monitoring and incident assessment
S3 Audit, QA, and Hospital Handoff Layer	S1 evidence, S2 alarms, operator action, QA decision, QR passport	Audit log, corrective record, validated action, explicit S3 audit record, exportable handoff-ready dossier	Human-in-the-loop accountability and handoff preparation
Integrity Layer	Canonical passport payload, QR/batch artifact	keccak256 hash, local anchor, optional external verification	Complementary audit evidence

Table S3: Detailed summary of the local AI-assisted evaluation. The *Success* and *Fail* columns denote successful/failed cases for the S1 rows and controlled responses/unhandled exceptions for S2. The 100-query S1 bank is partitioned by execution route: 36 queries resolve through the deterministic S1-1 field-lookup path and 64 through the generative S1-2 path, so the two route rows are disjoint and the pooled row reports the whole bank. S2 corresponds to the deterministic IoT alarm agent.

Component	**Evaluation protocol**	**Test cases**	**Success**	**Fail**	**Main result**
S1-1 Analytical Retrieval	Raw keyword exact-match over its deterministic field-lookup questions	36	34	2	94.4% raw; 36/36 (100%) under normalized numeric matching
S1-2 Generative Explanation	Three-evaluator panel on factual fidelity against the retrieved evidence package	64	64	0	100% factual fidelity ($\kappa = 0.91$)
S1 pooled (S1-1 + S1-2)	Raw keyword exact-match over the full question bank	100	98	2	98% raw; 100% under normalized numeric matching
S2 Operational Monitoring (field)	Deterministic state re-derived independently vs. deployed classifier on real telemetry	55,266	55,266	0	100% policy concordance (0 discrepancies)
S2 Operational Monitoring (robustness probe)	Deterministic state classification under nominal and forced edge cases	100	96	4	96% controlled responses; 4 input-validation exceptions
S2 infrastructure cases	Explicit infrastructure-event tests	3	3	0	Gateway and broker offline cases mapped to high-severity ALARM

top-K chunks ($K = 10$), collapsed them to a ranked list of unique documents (best rank per document), and scored the ranking against a document-level ground truth using Hit@k, mean reciprocal rank (MRR), and normalized discounted cumulative gain (nDCG@K) under binary relevance.

We deliberately separate two ground-truth regimes, mirroring the S1-1/S1-2 construct-validity split used throughout this work. For the **genetics** questions the relevant document is unambiguous and derived deterministically from the variety name and requested metric (e.g., a THC query for a given variety maps to `validated_info/{VARIETY}_Cannabinoides.md`); this is an *exact* ground truth. For the **operational** questions the relevant document is assigned by a lexical-overlap heuristic and then expert-revised; this is an *approximate* ground truth and is reported separately, never pooled with the exact subset. This approximate regime carries an important coverage limitation that we state explicitly because it depresses the reported figures: the heuristic assigned no relevant document at all to 38 of the 58 operational questions, and the scoring treats an empty relevance set as a miss, contributing zero to every metric. Inspection confirms that this is a labelling gap rather than a retrieval failure—for example, the query on clone-propagation procedure returns `SKILL_clonacion.md` at rank 1, a clearly pertinent document that carries no label. We therefore report the operational subset twice: over all 58 questions, which is the conservative figure a reader should use for the subset as published, and over the 20 questions that actually carry a label, which is the closest available estimate of retriever behaviour where ground truth exists. Completing the operational relevance judgements is required before these figures can be read as retrieval quality, and is left to future work. We stress that in production the genetics queries are resolved by the deterministic S1-1 route, which does not invoke the vector retriever; the genetics figures below therefore characterize what the retriever *would* return for those queries (a component diagnostic), not the accuracy of the S1-1 path.

These metrics address a specific reviewer concern: the main-text correctness figures for S1 are computed with a keyword/normalized heuristic on the final answer, whereas retrieval metrics isolate the quality of the evidence-selection

Table S4: Document-level retrieval metrics for the S1 vector index over the 100-question evaluation bank, computed with `nomic-embed-text` embeddings at retrieval depth $K = 10$. The genetics subset uses an exact document-level ground truth; the operational subset uses an expert-revised heuristic ground truth and is reported separately, both over all 58 questions and over the 20 that carry a non-empty relevance label—the heuristic left 38 unlabelled, and unlabelled questions score zero on every metric, so the all-questions row is a lower bound on retriever behaviour rather than a measurement of it. Binary relevance; nDCG normalized against the ideal ranking. Figures are generated by `eval_retrieval_metrics.mjs` and are not pooled across ground-truth regimes.

Subset	n	**Hit@1**	**Hit@3**	**Hit@5**	**MRR**	**nDCG@K**
Genetics (exact GT)	42	90.5%	100.0%	100.0%	0.95	0.96
Operational (heuristic GT, all)	58	13.8%	20.7%	25.9%	0.19	0.21
Operational (heuristic GT, labelled only)	20	40.0%	60.0%	75.0%	0.55	0.61

stage that precedes generation. Reporting them separately for the exact and heuristic ground-truth regimes keeps the interpretation honest: the exact regime bounds retriever behaviour on well-posed factual lookups, while the heuristic regime is indicative only and sensitive to the expert revision of the ground-truth annotations in `qrels_retrieval.json`.

4.2. S2 Deterministic Validation on Real Field Telemetry

The main-text S2 result is validated on real operational telemetry rather than on synthetic events alone. We independently re-derived the expected operational state (OK/WARNING/ALARM) from the stored values and thresholds of recorded anomaly objects, using Python (`validate_s2_real.py`) to check band labels and reconstruct severity from the logged anomaly list, with combination labels reusedand compared this re-derivation against the state the deployed S2 agent actually recorded for every historical evaluation stored in the traceability database (`local_iot/iot.db`). Table S5 reports the outcome: over 55,266 real evaluations spanning three active grow rooms and approximately eleven weeks of continuous operation (2026-03-29 to 2026-06-16), the deployed classifier matched the independent re-derivation with 100% concordance at both the per-metric banding level and the aggregated room-state level,

with zero discrepancies. Median wall-clock latency was 0.24 ms per evaluation (p90 0.40 ms). This validates the deterministic fidelity and reproducibility of the S2 engine against its published policy on real field data; it does not adjudicate the agronomic correctness of the threshold bands themselves, which would require expert-defined ground truth and is left to future work.

Table S5: S2 validation on real field telemetry (`local_iot/iot.db`, agent evaluations logged 2026-03-29 to 2026-06-16). Concordance between an independent Python re-derivation of the published alarm policy and the state recorded by the deployed S2 agent. Each logged evaluation stores its own metric value and the thresholds in force at decision time, so the re-derivation is self-contained and does not depend on the raw sensor-reading table, which retains only a rolling three-week window (8,463 readings, 2026-05-26 to 2026-06-16). Freshness states (284 OFFLINE) are excluded from the aggregated room-state check because they encode staleness rather than a banding decision.

Validation level	**Decisions**	**Discrepancies**	**Concordance**
Recorded anomaly-band classification	136,005	0	100.0%
Aggregated room state (OK/WARNING/ALARM)	54,982	0	100.0%

State distribution (real): 36,556 ALARM, 17,753 WARNING, 673 OK

Rooms: Vegetative 27,988; Mothers 13,748; Clones 13,529

Latency: median 0.24 ms/eval (p90 0.40 ms; $n = 100$ timed)

Table S6: S2 operational monitoring state distribution in the complementary 100-interaction robustness probe (synthetic). Nominal tests correspond to interactions 1–50 and forced edge-case tests to interactions 51–100.

State	**All tests**	**Nominal (1–50)**	**Edge (51–100)**
OK	60	37	23
WARNING	10	10	0
ALARM	26	3	23
STALE	0	0	0
OFFLINE	0	0	0
CRITICAL_ERROR	4	0	4

Table S7: Explicit infrastructure-event test cases for S2 operational monitoring.

Scenario	**Room**	**Expected state**	**Expected severity**
Gateway Offline	Cloning Room	ALARM	high
Broker Offline	Mother Plant Room	ALARM	high
Operational	Vegetative Growth Room	OFFLINE	medium

Table S8: Representative IoT operational thresholds (S2 policy layer). Target, Warning, and Alarm columns are shown as min / max.

Metric	**Unit**	**Target**	**Warning**	**Alarm**
Cloning Room				
Ambient Temp	°C	20 / 25	21 / 26	20 / 28
Relative Humidity	%	75 / 85	70 / 90	65 / 92
VPD	kPa	0.4 / 0.8	0.3 / 0.9	0.2 / 1.1
Flowering Room				
Ambient Temp	°C	22 / 26	20 / 28	18 / 30
Relative Humidity	%	45 / 60	40 / 65	35 / 70
VPD	kPa	1.0 / 1.4	0.8 / 1.6	0.6 / 1.8

5. Prototype Web Interface

This section describes the design of the PLANT2CLINIC-Agronomy web application and illustrates its interface (Figure S1).

Design rationale.. The web application is the single operator-facing surface of the edge node. Its design follows the same three principles that govern the platform as a whole (Section 10): it must remain fully usable without an active internet connection (*Edge-First*); it must keep probabilistic language assistance strictly separated from deterministic operational control (*Deterministic Con-*

Table S9: Functional validation checklist for the S3 audit, QA, and handoff workflow.

S3 workflow item	**Status**
Link S1 evidence to QR-linked record	Verified
Link S2 alarm to audit record	Verified
Register operator action	Verified
Register QA decision	Verified
Generate explicit S3 audit record	Verified
Export handoff-ready dossier	Verified
Verify record integrity via hash re-calculation	Verified
Hospital reception or patient data evaluation	Not evaluated

trol); and every state-changing action it exposes must be recorded together with its operational context for later verification (*Audit-by-Design*). Accordingly, the interface is rendered locally on the edge-computing node and communicates only with the local backend (Node.js/Express over a RESTful API), the filesystem-based traceability mirror and separate local IoT and integrity stores, and the local inference and messaging services (Ollama, MQTT); no view depends on external connectivity.

Module structure and navigation.. The interface is organised as a unified dashboard partitioned into four functional modules, each mapped to a distinct stage of the plant-to-clinic agronomic workflow and to the underlying agent services:

- **(A) Agronomy Module** — genetic registry, lot tracking, and QR-passport generation. This module is the entry point for traceability: it creates and links batch nodes (genetics, mother plants, clones, vegetative and flowering lots) and materialises the parent–child lineage that the integrity layer later verifies.
- **(B) Chemistry & Quality Module** — ethanol extraction, concentrate storage, and quality validations. It captures the transformation and QA

Table S10: Example of S1-S2-S3 evidence consolidation for a harvest event.

Evidence Type	**Content / Source**
S1 Knowledge	"The batch was harvested following SOP-AGRO-09. All visual inspections for Botrytis were negative." (Source: SOP-AGRO-09, Harvesting Protocol)
S2 Monitoring	"Room temperature remained within target range (22–26°C) for 98% of the flowering cycle. One minor humidity warning acknowledged at Day 45."
S3 Audit	"Batch released for Extraction by QA Lead on 2024-04-26. Integrity hash verified and anchored locally."

steps that convert plant material into concentrates, attaching analytical results and release/rejection decisions to the corresponding batch nodes.

- **(C) IoT Environmental Monitoring** — real-time telemetry, S2 operational alerts, and actuator control. This module surfaces the deterministic S2 monitoring agent: it displays room-level sensor readings, renders OK/WARNING/ALARM states through the fixed operational policy, and exposes bounded actuator actions.

- **(D) AI Evidence Assistant & Audit Layer** — the S1 conversational evidence assistant, the S3 audit and QA review panel, and integrity notarisation. This module co-locates the advisory and the audit surfaces, making the advisory/deterministic boundary visible to the operator within a single view.

View–agent mapping and the advisory/deterministic boundary.. The interface deliberately renders the three agent roles differently so that operators cannot mistake advisory output for an operational record. The S1 assistant (module D) is presented as a chat surface whose responses are grounded, retrieval-augmented

suggestions and are never written back to the traceability database as authoritative facts. The S2 agent (module C) is presented as a status-and-alert surface driven by deterministic thresholds rather than by generated text, so that alarm severity and room state remain fully accountable. The S3 audit workflow (module D) is presented as a review-and-release panel in which a human operator confirms QA decisions and triggers integrity verification. This separation mirrors the Presentation/Agent-Logic decoupling of the software architecture (Section 10): background monitoring and evidence generation run independently of the UI, and the UI exposes deterministic controls and human gates for every decision that alters operational state.

6. Data Integrity Audit Protocol

The following protocol describes the procedure for verifying the integrity of operational records within the PLANT2CLINIC-Agronomy platform. This protocol is intended for internal or external auditors.

Audit procedure.. The audit follows a "Trust-but-Verify" model based on cryptographic hashes:

1. **Sample selection**: The auditor selects a QR code or batch ID for verification.
2. **Verification request**: The auditor triggers the "Verify Integrity" function in the platform.
3. **Hash re-calculation**: The system reconstructs the canonical payload of the current record and re-calculates its keccak256 digest over the canonicalized payload; the same digest is used whether the original anchor was recorded locally or on an EVM-compatible network.
4. **Comparison**: The system compares the current hash against the original "notarized" hash stored in the local append-only integrity log. This log is append-only by application policy rather than cryptographically immutable, so it makes alteration detectable rather than impossible.

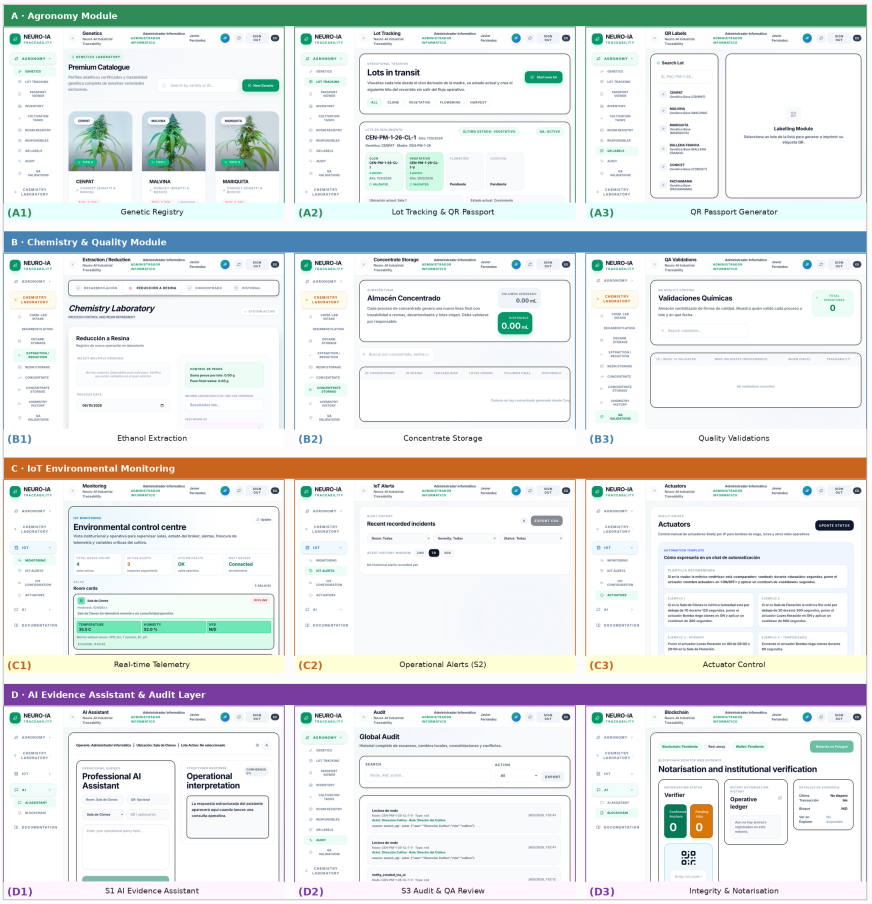


Figure S1: Representative screenshots of the PLANT2CLINIC-Agronomy web interface, illustrating the four main functional modules. **(A) Agronomy Module**: genetic registry, lot tracking with QR passport, and QR passport generator. **(B) Chemistry & Quality Module**: ethanol extraction, concentrate storage, and quality validations. **(C) IoT Environmental Monitoring**: real-time telemetry, S2 operational alerts, and actuator control. **(D) AI Evidence Assistant & Audit Layer**: S1 AI evidence assistant chat interface, S3 audit and QA review panel, and integrity notarisation. All interface views are rendered locally on the edge-computing node without external connectivity.

Verification pillars.. Each integrity check validates four critical pillars:

- **Node Data**: Batch ID, type, timestamps, and operational metadata.

- **Lineage**: The explicit parent-child relationship to ensure origin records are intact.
- **IoT Context**: Sensor readings (min, max, average) associated with the record.
- **QA Status**: Release/rejection decisions and registered auditor comments.

Interpretation of results..

- **Match**: Current data is identical to the notarized state. Integrity is confirmed.
- **Mismatch**: Data has been manually altered in the operational database after notarization. Critical finding requiring investigation.
- **No Record**: The record was not notarized or belongs to a non-critical stage.

6.1. Integrity-Layer Tampering Benchmark (S3)

The audit protocol of Section 6 specifies how a record is verified, but it does not by itself establish *which* classes of alteration the mechanism actually detects. We therefore executed a tampering benchmark directly against the production integrity module (`server/blockchain.js`), initialised on a temporary anchor store so that no production anchor was written. No component was re-implemented for the test: the benchmark calls the same canonicalization, digest and verification functions used by the platform, whose digest is `keccak256(toUtf8Bytes(canonicalJSON))`.

The benchmark comprises 36 cases built on a realistic traceability payload (a full mother–clone–vegetative lineage with IoT context and QA status). Of these, 31 are alterations that *must* be detected and 5 are semantic equivalents that must *not* be flagged. Reporting both classes matters: a mechanism that flags every change, including changes that preserve meaning, would be unusable in an audit workflow because every legitimate re-serialization would raise a false finding.

Table S11: Detection of tampering families by the S3 integrity layer. Counts are alterations that must be detected; semantically equivalent transformations are reported separately in the text. The anchor-store rewrite row is a deliberate boundary case (see below), and excluding it the detection rate is 29/29.

Family	**Alteration applied**	**Detected**
Field tampering	Single-field mutation of QR identifier, batch stage, THC/CBD value, room, timestamp, operator, and lineage parent	13/13
Deletion	Removal of required keys and of whole nested objects (IoT context, QA status, lineage)	8/8
Reordering	Array-element reordering within lineage and IoT reading lists	3/3
Replay	Re-submission of an already anchored payload under a different source reference	2/2
Concurrency	Interleaved anchoring of two payloads sharing one source reference	3/3
Anchor-store rewrite	Adversary with write access to the local anchor file recomputes the digest and rewrites the stored row	1/2
Total		**30/31**
Total excluding anchor-store rewrite		**29/29**

All 5 semantic equivalents behaved as specified (5/5): permuting object keys does not change the digest, because canonicalization sorts keys before serialization; and re-writing a numeric value in a different but equivalent notation (`2.23` $\rightarrow$ `2.230`) is likewise absorbed by normalization. Fig. S2 summarises both classes.

Declared boundary of the mechanism.. One case is designed to fail and is reported as a failure rather than removed. An adversary holding write access to the local anchor file can recompute the digest of a forged payload with the same public hash function and overwrite the stored row; verification then reports a

match. This is the direct consequence of anchors being co-located with the data they protect, and it delimits the claim made in this paper: the mechanism is tamper-*evident* against alteration of operational records, not tamper-*proof* against an attacker who controls the host. External anchoring exists precisely to remove this co-location, and is demonstrated here only as a non-production proof-of-concept. Two replay cases are also reported with a qualification: verification matches a payload against the set of anchored digests and does not impose a temporal order, so a superseded version of a record continues to verify after a legitimate update because both versions remain anchored. Ordering must be recovered from the audit timeline, not from the digest.

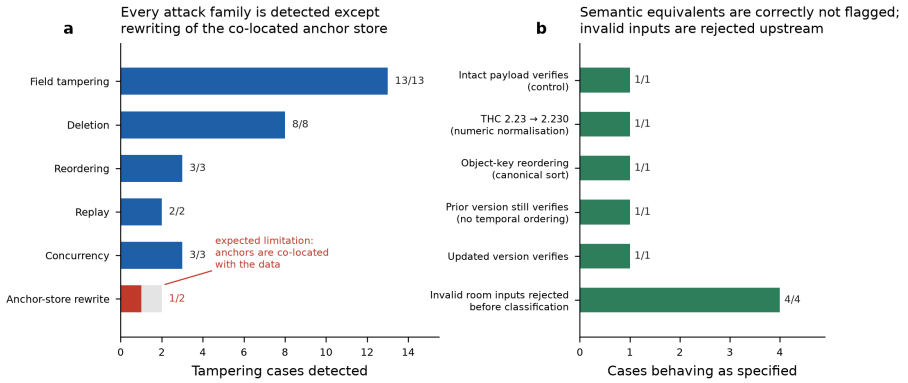


Figure S2: Tampering benchmark of the S3 integrity layer, executed against the production module. **(a)** Alterations detected per family; the only shortfall is the deliberate boundary case in which the co-located anchor store is itself rewritten. **(b)** Semantically equivalent transformations are correctly not flagged, and invalid room identifiers are rejected before any state classification is attempted.

Characterization of the four unhandled robustness cases.. The controlled robustness probe reported in the main text recorded 4 of 100 cases as `CRITICAL_ERROR` (Table S6). We characterized these directly against the live policy engine, driving it with a temporary copy of the field database. All four are invalid *inputs* rather than misclassifications: a non-existent room name, an empty string, a null value, and a numeric identifier. In all four the engine re-

solved no room and rejected the request *before* any state was computed, and in none of them was an alarm state emitted (0/4 emitted a state); the valid-room control resolved normally. The defect is therefore the *form* of the rejection — a generic exception instead of a typed error code, which is why the harness scored them as critical errors — and not the classification logic, which never had the opportunity to run. We report the original 96/100 figure unchanged rather than re-scoring it, and note that returning a typed validation error is a straightforward correction that we have deliberately left for a maintenance release rather than applying it to the code base evaluated here.

7. IoT Operational Policies

The S2 agent evaluates operational telemetry against a deterministic set of policies. Supplementary Table S6 shows representative thresholds used for the Cloning Room and Flowering Room.

8. Traceability Node Data Schema

The integrity layer hashes a canonicalized JSON payload representing the state of a batch. A simplified schema of the hashed artifact is shown below:

```
{
  "qr_id": "BATCH-2024-001",
  "type": "FLOWERING_BATCH",
  "parent_id": "BATCH-2024-VEG-001",
  "metadata": {
    "strain": "CBD-Rich-V1",
    "stage": "flowering",
    "started_at": "2024-04-10T08:00:00Z"
  },
  "lineage": ["GEN-001", "MOTHER-02", "CLONE-105"],
  "iot_context": {
```

```
    "avg_temp": 24.5,
    "avg_hum": 52.1,
    "last_alarm": "NONE"
  },
  "schema_version": "1.0"
}
```


9. Representative Evidence Consolidation

Supplementary Table S8 shows how the S3 layer consolidates heterogeneous evidence from S1 and S2 into an auditable record for the final batch dossier.

10. Software Design and Development

The PLANT2CLINIC-Agronomy platform was developed as a research prototype focusing on edge deployment and local-first autonomy.

Technology stack..

- **Backend**: Node.js with Express framework for the RESTful API.
- **Data Storage**: Filesystem snapshots and history for central traceability; separate SQLite databases for IoT and integrity services. The Health2Clinic demo has its own SQLite database.
- **IoT Integration**: MQTT (Eclipse Mosquitto) for asynchronous message passing between sensors and the S2 monitoring agent.
- **AI Orchestration**: Local Llama3:8B via the Ollama runtime, integrated into the S1 knowledge retrieval pipeline.
- **Frontend**: The current central and farmer applications use React/Vite; Health2Clinic is a separate browser–server application. No browser rendering benchmark is inferred from backend timings.

Architectural layers.. The software follows a modular design divided into four primary layers:

1. **Ingestion Layer**: Handles MQTT telemetry and barcode/QR scanning events.
2. **Persistence Layer**: Manages the Mirror Database (traceability) and the Integrity Database (notarization log).
3. **Agent Logic Layer**: Contains the S1, S2, and S3 services. This layer is decoupled from the UI to allow for independent execution of background monitoring and evidence generation.
4. **Presentation Layer**: Provides a unified dashboard for operational management, evidence review, and audit verification.

Development principles.. The development was guided by the following principles:

- **Edge-First**: All critical functions must operate without an active internet connection.
- **Deterministic Control**: AI components (S1) are advisory; all operational decisions (S2/S3) follow deterministic policies and human gates.
- **Audit-by-Design**: Every state change is recorded with its associated context (IoT, actor, timestamp) and is eligible for cryptographic anchoring.

11. S1 Agent (RAG Assistant) Full Evaluation Set

*Note: The PLANT2CLINIC platform operates in Spanish. The queries (**Question**), evaluation keywords (**Expected Key**), and verbatim system outputs (**Response Extract**) in this table were translated into English for this article; the original Spanish records are distributed verbatim in the machine-readable export described below.*

Provenance and scoring convention. This table is generated programmatically from the single master result file of the deployment run (`resultados_s1_camino_a.json`, run of 23 July 2026) by the script `regenerar_tabla_s1.py`, which aborts unless the per-row outcomes it publishes reproduce the aggregate figures reported in the main text. The table therefore cannot drift from the body of the article. Two independent outcome columns are reported for every query, because they answer different questions. **Raw** is exact string containment of the expected key in the emitted answer. **Norm.** is numeric-equivalence matching, which accepts the same quantity written in a different surface form. The two rows where the columns disagree (#22, #23) are format mismatches, not retrieval failures: the system returned the correct value `6.4%` against an expected key literally written `6.40`, as the **Response Extract** column shows. We publish both columns rather than only the favourable one so that the reader can verify this claim per row instead of taking it on trust. The same records are also released as `s1_query_level_results.csv` (one row per query, with the original Spanish text, both outcome flags, and per-query latency) so that the evaluation can be re-scored under a different criterion without re-running the system.

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
1	Genetics – THC	S1-1	What percentage of THC does the CENPAT variety have?	2.23	PASS	PASS	Technical information for the **CENPAT** variety: THC: **2.23%**.
2	Genetics – THC	S1-1	I need to know the THC level of CENPAT.	2.23	PASS	PASS	Technical information for the variety **CENPAT**: THC: **2.23%**.
3	Genetics – CBD	S1-1	How much CBD does the CENPAT plant have?	6.75	PASS	PASS	Technical information for the **CENPAT** variety: CBD: **6.75%**.
4	Genetics – CBD	S1-1	Exact CBD percentage for CENPAT.	6.75	PASS	PASS	Technical information for the **CENPAT** variety: CBD: **6.75%**.
5	Genetics – terpenes	S1-1	What is the main terpene in CENPAT?	Myrceno	PASS	PASS	Technical information for the **CENPAT** variety: Prominent terpenes: Myrceno (12.5%), Caryophylleno (8.2%), Pineno (5.4%).
6	Genetics – terpenes	S1-1	Tell me the outstanding terpenes of CENPAT.	Myrceno	PASS	PASS	Technical information for the **CENPAT** variety: Outstanding terpenes: Myrceno (12.5%), Caryophylleno (8.2%), Pineno (5.4%).

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
7	Genetics – documentary	S1-2	What validated documents exist for the CENPAT genetics?	CENPAT	PASS	PASS	The validated documentation for the CENPAT genetics includes: 1. [CENPAT_RNCP.md](validated_info/CENPAT_RNCP.md): Contains no information on THC, CBD or terpenes. 2. [CENPAT_Terpenos.md](validated_info/CENPAT_Terpenos.md): Contains no
8	Genetics – THC	S1-1	What percentage of THC does the MALVINA variety have?	5.76	PASS	PASS	Technical information for the **MALVINA** variety: THC: **5.76%**.
9	Genetics – THC	S1-1	I need to know the THC level of MALVINA.	5.76	PASS	PASS	Technical information for the **MALVINA** variety: THC: **5.76%**.
10	Genetics – CBD	S1-1	How much CBD does the MALVINA plant have?	0	PASS	PASS	Technical information for the **MALVINA** variety: CBD: **0%**.

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
11	Genetics – CBD	S1-1	Exact CBD percentage for MALVINA.	0	PASS	PASS	Technical information for the **MALVINA** variety: CBD: **0%**.
12	Genetics – terpenes	S1-1	What is the main terpene in MALVINA?	Limoneno	PASS	PASS	Technical information for the MALVINA variety: Notable terpenes: Limoneno (10.1%), Pineno (4.5%), Humuleno (3.2%).
13	Genetics – terpenes	S1-1	Tell me the notable terpenes of MALVINA.	Limoneno	PASS	PASS	Technical information for the **MALVINA** variety: Notable terpenes: Limoneno (10.1%), Pineno (4.5%), Humuleno (3.2%).
14	Genetics – documentary	S1-2	What validated documents exist for the MALVINA genetics?	MALVINA	PASS	PASS	The validated documentation available for MALVINA genetics includes: 1. validated_info/ MALVINA_Terpenos.md: Contains no information on terpenes. 2. validated_info/MALVINA_ RNCP.md: No RNCP document has been located for M

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
15	Genetics – THC	S1-1	What percentage of THC does the MARIQUITA variety have?	2.41	PASS	PASS	Technical information for the **MARIQUITA** variety: THC: **2.41%**.
16	Genetics – THC	S1-1	I need to know the THC level of MARIQUITA.	2.41	PASS	PASS	Technical information for the MARIQUITA variety: THC: **2.41%**.
17	Genetics – CBD	S1-1	How much CBD does the MARIQUITA plant have?	6.69	PASS	PASS	Technical information for the **MARIQUITA** variety: CBD: **6.69%**.
18	Genetics – CBD	S1-1	Exact CBD percentage for MARIQUITA.	6.69	PASS	PASS	Technical information for the **MARIQUITA** variety: CBD: **6.69%**.
19	Genetics – terpenes	S1-1	What is the main terpene in MARIQUITA?	Bisabolol	PASS	PASS	Technical information for the **MARIQUITA** variety: Notable terpenes: Bisabolol (15.2%), Myrceno (12.1%), Linalool (4.8%).
20	Genetics – terpenes	S1-1	Tell me the standout terpenes of MARIQUITA.	Bisabolol	PASS	PASS	Technical information for the **MARIQUITA** variety: Standout terpenes: Bisabolol (15.2%), Myrceno (12.1%), Linalool (4.8%).

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
21	Genetics – documentary	S1-2	What validated documents exist for the MARIQUITA genetics?	MARIQUITA	PASS	PASS	The MARIQUITA genetics has validated documentation in the following documents: 1. validated_ info/MARIQUITA_Terpenos.md: Validated Terpenes - MARIQUITA 2. validated_info/MARIQUITA_ RNCP.md: Validated RNCP - MARIQUITA 3. validated_
22	Genetics – THC	S1-1	What percentage of THC does the BALLENA FRANCA variety have?	6.40	**FAIL**	PASS	Technical information for the BALLENA FRANCA variety: THC: **6.4%**.
23	Genetics – THC	S1-1	I need to know the THC level of BALLENA FRANCA.	6.40	**FAIL**	PASS	Technical information for the BALLENA FRANCA variety: THC: **6.4%**.
24	Genetics – CBD	S1-1	How much CBD does the BALLENA FRANCA plant have?	6.55	PASS	PASS	Technical information for the **BALLENA FRANCA** variety: CBD: **6.55%**.
25	Genetics – CBD	S1-1	Exact CBD percentage for BALLENA FRANCA.	6.55	PASS	PASS	Technical information for the BALLENA FRANCA variety: CBD: **6.55%**.

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
26	Genetics – terpenes	S1-1	What is the main terpene in BALLENA FRANCA?	Limoneno	PASS	PASS	Technical information for the BALLENA FRANCA variety: Prominent terpenes: Limoneno (18.5%), Caryophylleno (10.2%), Myrceno (7.4%).
27	Genetics – terpenes	S1-1	Tell me the outstanding terpenes of BALLENA FRANCA.	Limoneno	PASS	PASS	Technical information for the variety **BALLENA FRANCA**: Outstanding terpenes: Limoneno (18.5%), Caryophylleno (10.2%), Myrceno (7.4%).
28	Genetics – documentary	S1-2	What validated documents exist for the BALLENA FRANCA genetics?	BALLENA FRANCA	PASS	PASS	The validated documentation for the BALLENA FRANCA genetics includes: 1. Validated RNCP: Contains no specific information on THC, CBD and terpenes. 2. Validated Terpenes: Contains no specific information on THC, CBD and terpenes

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
29	Genetics – THC	S1-1	What percentage of THC does the CONICET variety have?	8.31	PASS	PASS	Technical information for the **CONICET** variety: THC: **8.31%**.
30	Genetics – THC	S1-1	I need to know the THC level of CONICET.	8.31	PASS	PASS	Technical information for the CONICET variety: THC: 8.31%.
31	Genetics – CBD	S1-1	How much CBD does the CONICET plant have?	0.05	PASS	PASS	Technical information for the **CONICET** variety: CBD: **0.05%**.
32	Genetics – CBD	S1-1	Exact CBD percentage for CONICET.	0.05	PASS	PASS	Technical information for the **CONICET** variety: CBD: **0.05%**.
33	Genetics – terpenes	S1-1	What is the main terpene in CONICET?	Pineno	PASS	PASS	Technical information for the **CONICET** variety: Notable terpenes: Pinene (20.1%), Terpinolene (12.5%), Ocimene (8.2%).

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
34	Genetics – terpenes	S1-1	Tell me the prominent terpenes of CONICET.	Pineno	PASS	PASS	Technical information for the **CONICET** variety: Prominent terpenes: Pineno (20.1%), Terpinoleno (12.5%), Ocimeno (8.2%).
35	Genetics – documentary	S1-2	What validated documents exist for the CONICET genetics?	CONICET	PASS	PASS	There are no validated documents containing specific information on the CONICET genetics regarding THC, CBD, and terpenes. However, two validated documents related to the CONICET genetics are found: * [2] validated
36	Genetics – THC	S1-1	What percentage of THC does the PACHAMAMA variety have?	0.96	PASS	PASS	Technical information for the **PACHAMAMA** variety: THC: **0.96%**.
37	Genetics – THC	S1-1	I need to know the THC level of PACHAMAMA.	0.96	PASS	PASS	Technical information for the **PACHAMAMA** variety: THC: **0.96%**.

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
38	Genetics – CBD	S1-1	How much CBD does the PACHAMAMA plant contain?	16.09	PASS	PASS	Technical information for the PACHAMAMA variety: CBD: **16.09%**.
39	Genetics – CBD	S1-1	Exact CBD percentage for PACHAMAMA.	16.09	PASS	PASS	Technical information for the **PACHAMAMA** variety: CBD: **16.09%**.
40	Genetics – terpenes	S1-1	What is the main terpene in PACHAMAMA?	Bisabolol	PASS	PASS	Technical information for the **PACHAMAMA** variety: Notable terpenes: Bisabolol (28.9%), Caryophylleno (16.9%), Myrceno (13%), Alfa-Pineno (5.5%), alfa-Humuleno (5.5%).
41	Genetics – terpenes	S1-1	Tell me the notable terpenes of PACHAMAMA.	Bisabolol	PASS	PASS	Technical information for the **PACHAMAMA** variety: Notable terpenes: Bisabolol (28.9%), Caryophylleno (16.9%), Myrceno (13%), Alfa-Pineno (5.5%), alfa-Humuleno (5.5%).

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
42	Genetics – documentary	S1-2	What validated documents exist for the PACHAMAMA genetics?	PACHAMAMA	PASS	PASS	The PACHAMAMA genetics has the following validated documents: 1. **PACHAMAMA_RNCP.md**: No information is available about the content of the original PDF due to insufficient extractable text. 2. **PACHAMAMA_Cannabin
43	SOP – clones	S1-2	At what temperature should the clone room be kept? Variant 1.	temperatura, clones	PASS	PASS	The optimal temperature for keeping the clone room is between 22°C and 26°C, according to the validated documentation [1].
44	SOP – drying	S1-2	What is the drying or curing process of the flowers like? Variant 1.	secado, curing, humedad	PASS	PASS	The drying or curing process of the flowers refers to the final phase of cannabis cultivation, where the aim is to achieve the optimal balance between humidity and temperature so that the flowers mature naturally. There is no in

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
45	SOP – humidity	S1-2	What should I do if there is a humidity spike during flowering? Variant 1.	humedad, floracion	PASS	PASS	**Technical answer** In the context of the question "What should I do if there is a humidity spike during flowering?", there is no specific information on terpenes, THC, or CBD that could help resolve this issue. However, according to
46	Biosafety	S1-2	What are the biosecurity measures upon entering the bunker? Variant 1.	bioseguridad, guantes	PASS	PASS	There are no biosecurity measures upon entering the bunker in the context provided. The document does not mention entry into a bunker or any security measure related to it.
47	Cultivation	S1-2	When is apical pruning performed? Variant 1.	poda, vegetativo	PASS	PASS	The specific timing of apical pruning is not mentioned in any of the documents provided. However, as S1 Chat-Agent, I can inform you that apical pruning is a common process in cannabis production

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
48	IoT – pH	S1-2	How is the pH sensor calibrated? Variant 1.	pH, calibr	PASS	PASS	Calibration of the pH sensor is performed according to the procedure established in the validated documentation [1]. In particular, it is indicated that the EC and pH probes must be maintained under a verification or calibration procedure s
49	IoT – EC	S1-2	What is the ideal EC level during flowering bulking/ fattening stage? Variant 1.	EC, floracion	PASS	PASS	There is no direct information on the ideal EC level in flowering bulking/ fattening in the provided documents. However, related information can be found re

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
50	RAG	S1-2	What is RAG and how does it analyze documents? Variant 1.	RAG, document	PASS	PASS	RAG (Recovery and Analysis Group) is an operational mode of the S1 assistant that focuses on document extraction and analysis. There are two modes of operation: strict extractive RAG: uses only document retrieval, without (Note: The Spanish excerpt cuts off mid-sentence; I've translated to the point where the text ends.)
51	Traceability	S1-2	How is the traceability passport of a batch constructed? Variant 1.	trazabilidad, lote	PASS	PASS	The construction of the traceability passport of a batch is carried out throughout the different phases of the process, from agronomy to pharmacy. Below are the key steps to construct the passport: 1

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
52	Audit	S1-2	What evidence must be kept for a GACP audit? Variant 1.	auditoria, evidencia	PASS	PASS	For a GACP audit, the evidence that must be kept includes: (1) Audit result (PASS/FAIL); (2) Non-conformities found and category; (3) Batch and cycle indicators; (4) Efficacy of previous corrective actions.
53	Validations	S1-2	When does a deviation require human review? Variant 1.	revision, humana	PASS	PASS	There is no specific information about deviation requiring human review in the provided context. However, it is recommended to verify the condition of clone room out of range for more than 4 hours according to documentation.

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
54	IoT	S1-2	What IoT parameters are critical in a cultivation room? Variant 1.	temperatura, humedad	PASS	PASS	The critical IoT parameters in a cultivation room are: Temperature (T), Humidity (H), VPD (vapor pressure deficit), EC (electrical conductivity), pH, DLI (daily light integral). These parameters are critical because they
55	Inventory	S1-2	How are inventory, lot, and room related? Variant 1.	inventario, lote	PASS	PASS	The relationship between inventory, lot, and room is described in the context as follows: A lot (‘lotes_ vegetativo‘, ‘lotes_floracion‘, etc.) can be associated with a specific IoT room. The location of a node (‘Location‘) changes

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
56	Blockchain	S1-2	What information is anchored in blockchain and what is it used for? Variant 1.	blockchain, ancla	PASS	PASS	The information anchored in blockchain and its purpose: The blockchain service adds a layer of cryptographic evidence to the PLANT2CLINIC Traceability system, certifying the integrity of specific passports or snapshots.
57	Clinical trial	S1-2	How is the dispensation to a trial patient documented? Variant 1.	dispensacion, paciente	PASS	PASS	Valid documentation for dispensation to a trial patient in variant 1 does not provide detailed information on the specific chemical components of the product. However, it can be inferred that the trial involves
58	Extraction	S1-2	What controls apply during resin extraction? Variant 1.	extraccion, resina	PASS	PASS	During resin extraction, the following controls are applied: 1. **PHASE 1 — Flower Reception:** * Verify before accepting: + Identifiable harvest lot QR code + Real weight measured and recorded + Tolerance of p

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
59	Storage	S1-2	How is concentrate storage controlled? Variant 1.	almacenamiento, concentrado	PASS	PASS	The answer to the question "How is concentrate storage controlled?" is found in variant 1 of the validated documentation. In phase 5, "Concentrate Preparation," the batches are mixed to create the concentr
60	Quality	S1-2	What steps must the quality team follow before releasing a batch? Variant 1.	calidad, lote	PASS	PASS	To release a batch, the quality team must follow the following steps: 1. Verify that all mandatory data are recorded. 2. Calculate the rooting rate and detect deviations that require action. 3. Record l
61	Alerts	S1-2	What immediate actions does the system recommend in response to a critical alarm? Variant 1.	alarma, accion	PASS	PASS	The recommended immediate action in response to a critical alarm (ALARM high or OFFLINE critical) is: * Immediate: The operator must press the "Treat" button from the interface and log the resolution note associated with the original alert.

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
62	Documents	S1-2	Where should validated information be uploaded so that the RAG can use it? Variant 1.	validated_info, document	PASS	PASS	To upload validated information for the RAG to use, you must place it in the ‘validated_info‘ folder and follow the established file formats and structure.
63	SOP – clones	S1-2	At what temperature should the clone room be maintained? Variant 2.	temperatura, clones	PASS	PASS	According to the validated documentation, to keep the clone room at an optimal temperature a range of 22-26°C is recommended.
64	SOP – drying	S1-2	What is the process of drying or curing of the flowers? Variant 2.	secado, curing, humedad	PASS	PASS	The process of drying or curing of the flowers is an important step in cannabis production. Below is a general description of the process: 1. **Identify the current phase of the plant**: Before beginning the

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
65	SOP – humidity	S1-2	What should I do if there is a humidity spike during flowering? Variant 2.	humedad, floracion	PASS	PASS	There is no specific information about terpenes in the context provided. However, as a chat agent, I can suggest some strategies to address a humidity spike during flowering: * Check the humidity levels
66	Biosafety	S1-2	What are the biosafety measures when entering the bunker? (Variant 2)	bioseguridad, guantes	PASS	PASS	There are no biosafety measures for entering the bunker in the provided context. The validated documentation does not mention anything about biosafety measures when entering the bunker.
67	Cultivation	S1-2	At what point is apical pruning performed? Variant 2.	poda, vegetativo	PASS	PASS	The specific point at which apical pruning is performed is not mentioned in the provided context. However, as an S1 Chat-Agent agent, I can inform you that apical pruning is a common process in cannabis production and others

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
68	IoT – pH	S1-2	How is the pH sensor calibrated? Variant 2.	pH, calibr	PASS	PASS	The calibration of the pH sensor is described in step 2 of the validated documentation: ### Step 2 — Reading and validation of data For each reading,
69	IoT – EC	S1-2	What is the ideal EC level during flowering fattening/ bulking stage? Variant 2.	EC, floracion	PASS	PASS	According to the context, the ideal EC (Electrical Conductivity) level during flowering fattening/bulking is not explicitly mentioned in the provided documents. However, in document [4] "SKILL_ cannabis-cultivation-guide.md",
70	RAG	S1-2	What is RAG and how does it analyze documents? Variant 2.	RAG, document	PASS	PASS	RAG (Recovery and Analysis Group) is an operating mode of the S1 assistant that focuses on document extraction and analysis. There are two variants: * Strict extractive RAG: uses only document retrieval, without intent

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
71	Traceability	S1-2	How is the traceability passport of a batch constructed? Variant 2.	trazabilidad, lote	PASS	PASS	The construction of the traceability passport of a batch is organized in 7 levels: 1. **Base Genetics** (‘genetica_base‘): contains basic genetic information, such as Genetic ID, Variety, Lineage, Notes, Image_ URL and Doc
72	Audit	S1-2	What evidence should be retained for a GACP audit? Variant 2.	auditoria, evidencia	PASS	PASS	For a GACP audit, the evidence that must be retained is: 1. Audit result (PASS/FAIL) 2. Non-conformities found and category 3. Batch and cycle indicators 4. Effectiveness of previous corrective actions In
73	Validations	S1-2	When does a deviation need human review? Variant 2.	revision, humana	PASS	PASS	There is no specific information about when a deviation needs human review in the context provided. However, it can be inferred that human review is necessary in certain cases, such as: * In variant 2, it men

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
74	IoT	S1-2	What IoT parameters are critical in a growing room? Variant 2.	temperatura, humedad	PASS	PASS	The critical IoT parameters in a growing room are: * Temperature (T) * Humidity (H) * Vapor pressure deficit (VPD) * Daily light integral (DLI) * Substrate temperature (T substrate) * Electrical conductivity (EC)
75	Inventory	S1-2	How are inventory, batch, and room related? Variant 2.	inventario, lote	PASS	PASS	The relationship between inventory, batch, and room in the context of phytopharmaceutical traceability is as follows: * A **batch** is a set of plants or botanical products that share specific characteristics, such as variety,
76	Blockchain	S1-2	What information is anchored on blockchain and what is it used for? Variant 2.	blockchain, ancla	PASS	PASS	According to the validated documentation, the blockchain service anchors information in the following way: * The batch passport is stored in a separate local database (‘local_blockchain/blockchain.db‘)

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
77	Clinical trial	S1-2	How is the dispensation to a trial patient documented? Variant 2.	dispensacion, paciente	PASS	PASS	The valid documentation does not provide specific information about dispensation to a trial patient in the context of PACHAMAMA. However, general information about the trial and dispensations can be found in
78	Extraction	S1-2	What controls are applied during resin extraction? Variant 2.	extraccion, resina	PASS	PASS	During resin extraction, the controls applied are: * **PHASE 3 — Resin Batch:** Consolidate data from all EX in the batch: + All ExtractoIDs listed + All ExtractoIDs have confirmed FTIR + Analy
79	Storage	S1-2	How is the storage of concentrates controlled? Variant 2.	almacenamiento, concentrado	PASS	PASS	Regarding the storage of concentrates, it is controlled through traceability and quality control processes. The following are the key steps: 1. **Trigger Concentrate Preparation**: The material is mixed with

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
80	Quality	S1-2	What steps should quality follow before releasing a batch? Variant 2.	calidad, lote	PASS	PASS	To release a batch before closing, the following steps must be followed: 1. Identified phase (Clonadora or Batch Registration) 2. All mandatory data recorded 3. Rooting rate calculated and compared with historical data
81	Alerts	S1-2	What immediate actions does the system recommend in the event of a critical alarm? Variant 2.	alarma, accion	PASS	PASS	The question is: "What immediate actions does the system recommend in the event of a critical alarm?" According to the validated documentation, in the event of a critical alarm (ALARM high),
82	Documents	S1-2	Where should validated information be uploaded so that RAG can use it? Variant 2.	validated_info, document	PASS	PASS	To upload validated information for RAG to use, it is recommended to use the validated documentation files provided in the "validated_info" folder.

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
83	SOP – clones	S1-2	At what temperature should the cloning room be maintained? Variant 3.	temperatura, clones	PASS	PASS	According to validated documentation, it is recommended to maintain the cloning room at a temperature between 22-26°C.
84	SOP – drying	S1-2	What is the drying or curing process of the flowers like? Variant 3.	secado, curing, humedad	PASS	PASS	The drying or curing process of the flowers is an important step in cannabis production. Next,
85	SOP – humidity	S1-2	What should I do if there is a humidity spike during flowering? Variant 3.	humedad, floracion	PASS	PASS	For a humidity spike during flowering, I recommend: • Verify the water content in the substrate and adjust as necessary. • Ensure that the irrigation system is functioning correctly and is not over-saturating.
86	Biosafety	S1-2	What are the biosafety measures when entering the bunker? Variant 3.	bioseguridad, guantes	PASS	PASS	There are no biosafety measures when entering the bunker in variant 3. The documentation does not mention anything about specific biosafety measures for entering the bunker. However, it can be inferred that the system has an appro

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
87	Cultivation	S1-2	At what point is apical pruning performed? Variant 3.	poda, vegetativo	PASS	PASS	There is no information in the context about the timing of apical pruning in variant 3. However, I can search the validated documentation for more information. In the documentation [1] docs/GUIA_ TEST_PWA_MO
88	IoT – pH	S1-2	How is the pH sensor calibrated? Variant 3.	pH, calibr	PASS	PASS	The pH sensor calibration is performed in Step 2 — Data reading and validation, within the plant observation process. In this step, it is verified that each reading is valid before analysis. For the parameter
89	IoT – EC	S1-2	What is the ideal EC level in flowering fattening? Variant 3.	EC, floracion	PASS	PASS	According to the validated documentation [1], there is no ideal EC (Electrical Conductivity) level specifically mentioned for the flowering fattening phase in variant 3.

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
90	RAG	S1-2	What is the RAG and how does it analyze documents? Variant 3.	RAG, document	PASS	PASS	The RAG (Recovery and Analysis Group) is an operating mode of the S1 assistant that focuses on document extraction and analysis.
91	Traceability	S1-2	How is the traceability passport of a batch constructed? Variant 3.	trazabilidad, lote	PASS	PASS	To construct the traceability passport of a batch, the following process is followed: 1. **Mother Plant Registry**: The genetics, germination date are documented and a unique QR code is assigned to the mother plant. 2. **Clo
92	Audit	S1-2	What evidence must be retained for a GACP audit? Variant 3.	auditoria, evidencia	PASS	PASS	For a GACP audit, the evidence that must be retained includes: audit result (PASS/FAIL), non-conformities found and category, batch and cycle indicators, and efficacy of previous corrective actions.

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
93	Validations	S1-2	When does a deviation require human review? Variant 3.	revision, humana	PASS	PASS	According to validated documentation [1] SKILL_ clonacion.md, a deviation requires human review when: the calculated rooting rate compared with historical data is not within the normal range. There are registered non-conformities
94	IoT	S1-2	What IoT parameters are critical in a cultivation room? Variant 3.	temperatura, humedad	PASS	PASS	Critical IoT parameters in a cultivation room are: Temperature (T), Humidity (H), VPD (Vapor Pressure Deficit), DLI (Daily Light Integral), Temperature of substrate (T substrate), EC (Electrical Conductivity), pH.

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
95	Inventory	S1-2	How are inventory, batch, and room related? Variant 3.	inventario, lote	PASS	PASS	The relationship between inventory, batch, and room is described in validated documentation as follows: A batch (‘lotes_vegetativo‘, ‘lotes_floracion‘, etc.) can have multiple clones (‘clones‘) that are located in a specific room.
96	Blockchain	S1-2	What information is anchored in blockchain and what is it used for? Variant 3.	blockchain, ancla	PASS	PASS	The information that is anchored in blockchain and its function is to certify the existence of a specific passport or snapshot with determined content on a specific date. The Blockchain service adds a layer of cryptographic evidence
97	Clinical trial	S1-2	How is the dispensation to a trial patient documented? Variant 3.	dispensacion, paciente	PASS	PASS	Valid documentation does not provide specific information on dispensation to a trial patient in the context of variant 3. However, I can respond conservatively and indicate that I cannot interpret the RNCP.

Continued on next page

Table S12 – continued from previous page

#	Category	Route	Question	Expected Key	Raw	Norm.	Response Extract
98	Extraction	S1-2	What controls are applied during resin extraction? Variant 3.	extraccion, resina	PASS	PASS	During resin extraction, the controls applied are: Loss $\leq$ 25%: continue; Record EtOH volume (L); Record IN and OUT time of first maceration; If second maceration occurs, record IN and OUT time.
99	Storage	S1-2	How is the storage of concentrates controlled? Variant 3.	almacenamiento, concentrado	PASS	PASS	To control the storage of concentrates, the following steps are recommended: 1. Calculate yields: • Decarboxylation yield: Loss % = (Input weight - Extraction mass) / Input weight $\times$ 100 • Yield
100	Quality	S1-2	What steps should quality follow before releasing a batch? Variant 3.	calidad, lote	PASS	PASS	To release a batch of clones, the following steps must be followed: 1. Verify that all mandatory data are registered. 2. Calculate the rooting rate and compare it with the historical data of the same cultivar. 3. No-c

12. S2 Agent (Operational Monitoring) Full Evaluation Set

#	Event	Temp	Hum	Status	Severity	Expectation	Result
1	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
2	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
3	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
4	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
5	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
6	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
7	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
8	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
9	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
10	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
11	telemetry	25.5	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
12	telemetry	27.0	50.0	WARNING	low	Normal	Flowering room in WARNING state due to temperature deviations.
13	telemetry	28.5	50.0	WARNING	low	Normal	Flowering room in WARNING state due to temperature deviations.
14	telemetry	30.0	50.0	WARNING	low	Normal	Flowering room in WARNING state due to temperature deviations.

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
15	telemetry	31.5	50.0	ALARM	high	Normal	Flowering room in ALARM state due to temperature deviations.
16	telemetry	30.0	50.0	WARNING	low	Normal	Flowering room in WARNING state due to temperature deviations.
17	telemetry	28.5	50.0	WARNING	low	Normal	Flowering room in WARNING state due to temperature deviations.
18	telemetry	27.0	50.0	WARNING	low	Normal	Flowering room in WARNING state due to temperature deviations.
19	telemetry	25.5	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
20	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
21	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
22	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
23	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
24	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
25	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
26	telemetry	24.0	55.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
27	telemetry	24.0	60.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
28	telemetry	24.0	65.0	WARNING	low	Normal	Flowering room in WARNING state due to humidity deviations.
29	telemetry	24.0	70.0	WARNING	low	Normal	Flowering room in WARNING state due to humidity deviations.
30	telemetry	24.0	75.0	ALARM	high	Normal	Flowering room in ALARM state due to humidity deviations.

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
31	telemetry	24.0	70.0	WARNING	low	Normal	Flowering room in WARNING state due to humidity deviations.
32	telemetry	24.0	65.0	WARNING	low	Normal	Flowering room in WARNING state due to humidity deviations.
33	telemetry	24.0	60.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
34	telemetry	24.0	55.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
35	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
36	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
37	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
38	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
39	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
40	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
41	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
42	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
43	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
44	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
45	gateway_offline	24.0	50.0	ALARM	high	Normal	Critical infrastructure incident: gateway_offline.
46	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
47	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
48	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
49	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
50	telemetry	24.0	50.0	OK	low	Normal	Flowering room in OK state with recent telemetry within operational parameters...
51	telemetry	Override	Override	OK	low	Data type failure (string)	Flowering room in OK state with recent telemetry within operational parameters...

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
52	telemetry	Override	Override	OK	low	Empty/null metrics	Flowering room in OK state with recent telemetry within operational parameters...
53	telemetry	Override	Override	ALARM	high	Extreme outlier (9999°C)	Flowering room in ALARM state due to temperature deviations.
54	telemetry	Override	Override	ALARM	high	Absolute zero (-273°C)	Flowering room in ALARM state due to temperature deviations.
55	telemetry	Override	Override	ALARM	high	Negative humidity	Flowering room in ALARM state due to humidity deviations.
56	telemetry	Override	Override	OK	low	NaN values	Flowering room in OK state with recent telemetry within operational parameters...
57	telemetry	Override	Override	OK	low	Incomplete payload (missing fields)	Flowering room in OK state with recent telemetry within operational parameters...
58	telemetry	Override	Override	OK	low	Unrecognized metrics	Flowering room in OK state with recent telemetry within operational parameters...
59	telemetry	Override	Override	OK	low	Empty metrics object	Flowering room in OK state with recent telemetry within operational parameters...

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
60	telemetry	24.0	50.0	OK	low	Metrics is undefined	Flowering room in OK state with recent telemetry within operational parameters...
61	telemetry	24.0	50.0	CRITICAL_ ERROR	high	Non-existent room	Exception: Invalid room
62	telemetry	24.0	50.0	CRITICAL_ ERROR	high	Empty room	Exception: Invalid room
63	telemetry	24.0	50.0	CRITICAL_ ERROR	high	Null room	Exception: Invalid room
64	telemetry	24.0	50.0	CRITICAL_ ERROR	high	Room number (12345)	Exception: Invalid room
65	alien_invasion	24.0	50.0	OK	low	Event de infraestructura desconocido	Flowering room in OK state with recent telemetry within operational parameters...
66	null	24.0	50.0	OK	low	Event de infraestructura null	Flowering room in OK state with recent telemetry within operational parameters...
67	telemetry	24.0	50.0	OK	low	Validate operation	Flowering room in OK state with recent telemetry within operational parameters...

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
68	telemetry	24.0	50.0	OK	low	Validate operation	Flowering room in OK state with recent telemetry within operational parameters...
69	telemetry	24.0	50.0	OK	low	Validate operation	Flowering room in OK state with recent telemetry within operational parameters...
70	telemetry	24.0	50.0	OK	low	Validate operation	Flowering room in OK state with recent telemetry within operational parameters...
71	telemetry	Override	Override	OK	low	Payload in String format	Flowering room in OK state with recent telemetry within operational parameters...
72	telemetry	Override	Override	OK	low	Payload Array instead of Object	Flowering room in OK state with recent telemetry within operational parameters...
73	telemetry	Override	Override	OK	low	Deeply nested payload	Flowering room in OK state with recent telemetry within operational parameters...
74	telemetry	24.0	50.0	OK	low	Validate operation	Flowering room in OK state with recent telemetry within operational parameters...

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
75	telemetry	24.0	50.0	OK	low	Validate operation	Flowering room in OK state with recent telemetry within operational parameters...
76	telemetry	24.0	50.0	OK	low	Validate operation	Flowering room in OK state with recent telemetry within operational parameters...
77	telemetry	24.0	50.0	OK	low	Validate operation	Flowering room in OK state with recent telemetry within operational parameters...
78	telemetry	24.0	50.0	OK	low	Validate operation	Flowering room in OK state with recent telemetry within operational parameters...
79	telemetry	24.0	50.0	OK	low	Validate operation	Flowering room in OK state with recent telemetry within operational parameters...
80	telemetry	24.0	50.0	OK	low	Validate operation	Flowering room in OK state with recent telemetry within operational parameters...
81	telemetry	10.0	10.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum deviations.

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
82	telemetry	35.0	90.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum/flowering deviations.
83	telemetry	10.0	10.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum deviations.
84	telemetry	35.0	90.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum/flowering deviations.
85	telemetry	10.0	10.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum deviations.
86	telemetry	35.0	90.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum/flowering deviations.
87	telemetry	10.0	10.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum deviations.
88	telemetry	35.0	90.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum/flowering deviations.
89	telemetry	10.0	10.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum deviations.

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
90	telemetry	35.0	90.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum/flowering deviations.
91	telemetry	10.0	10.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum deviations.
92	telemetry	35.0	90.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum/flowering deviations.
93	telemetry	10.0	10.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum deviations.
94	telemetry	35.0	90.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum/flowering deviations.
95	telemetry	10.0	10.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum deviations.
96	telemetry	35.0	90.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum/flowering deviations.
97	telemetry	10.0	10.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum deviations.

Continued on next page

Table S13 – continued from previous page

#	Event	Temp	Hum	Status	Severity	Expectation	Result
98	telemetry	35.0	90.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum/flowering deviations.
99	telemetry	10.0	10.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum deviations.
100	telemetry	35.0	90.0	ALARM	high	Stress: Constant sudden changes	Flowering room in ALARM state due to temp/hum/flowering deviations.

Scientific workflow of the prototype.. Fig. S3 shows how validated procedures, traceability, IoT context, deterministic policies, the S1 evidence-retrieval service, S2 operational monitoring, S3 audit validation and hospital handoff, and the experimental evaluation are connected. The figure makes explicit that the contribution is the controlled integration of these layers rather than any single module in isolation.

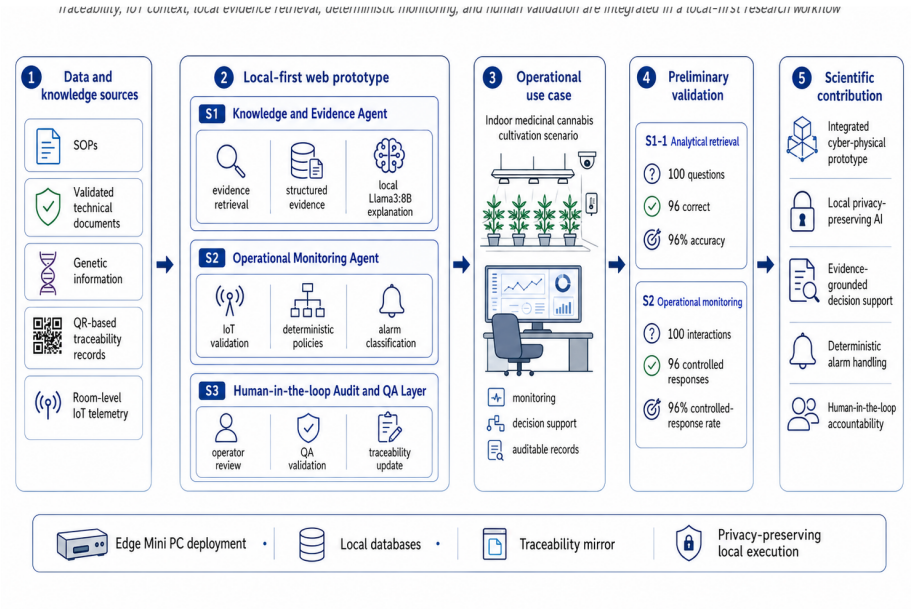


Figure S3: Scientific workflow of the proposed prototype. Data and knowledge sources are integrated into a local-first digital platform prototype composed of S1 evidence retrieval, S2 operational monitoring, and S3 human-in-the-loop audit. The workflow links the operational use case with preliminary validation and the scientific contribution of the system.

Abbreviations and Acronyms

This article combines vocabularies from three domains that do not usually appear together—pharmaceutical quality assurance, agricultural sensing, and retrieval-augmented language modelling—and each contributes its own abbreviations. Table S14 collects every abbreviation used in the main text and in this supplement, grouped by domain, so that a reader coming from one of the three fields can follow the passages written in the vocabulary of the other two. Two conventions are worth stating explicitly because they carry most of the notation load in the results. First, **S1**, **S2** and **S3** denote the three functional layers of the platform, and the suffixed forms **S1-1** and **S1-2** denote the two routes *within* S1—the deterministic analytical-retrieval route and the generative explanation route respectively—a distinction that matters throughout because the two routes have different accuracy, latency and failure modes and must not be pooled into a single figure. Second, tables and sections prefixed with **S** followed by a number (S3–S9) refer to the Supplementary Material, whereas S1–S3 without such a prefix always refer to the architectural layers; the surrounding text disambiguates, and the abbreviation table lists both senses.

Table S14: Abbreviations and acronyms used in the main text and supplement, grouped by domain.

Abbreviation	**Expansion and meaning in this article**
Platform architecture (this work)	
PLANT2CLINIC	Name of the platform reported here: a plant-to-clinic traceability and decision-support system for medicinal cannabis
S1	Functional layer for evidence extraction and explanation (RAG-based)
S1-1	Deterministic analytical-retrieval route within S1 (structured field lookup, no generation)

Table S14 (continued)

Abbreviation	Expansion and meaning in this article
S1-1a	Submode of S1-1: structured field lookup over typed traceability records by deterministic pattern matching; no embedding model and no language model participate
S1-1b	Submode of S1-1: vector retrieval over indexed SOP chunks with a non-generative neural embedding model (`nomic-embed-text`, cosine similarity)
S1-2	Generative explanation route within S1 (local LLM conditioned on retrieved evidence)
S2	Deterministic operational-monitoring layer (IoT alarm and state engine)
S2-IoT	S2 telemetry-analytics service (e.g. daily-light-integral estimation)
S2-E	S2 emergency-classification service
S3	Human-in-the-loop governance, QA audit and dossier-generation layer
R0–R5	The six evidence regimes of the abstention benchmark (control, empty, irrelevant, conflicting, poisoned value, injected instruction)
C-Trace	End-to-End Traceability Reconstructor, a clinical-phase agent (outside the scope of this article)
BIM	Blinding Integrity Monitor, a clinical-phase agent (outside the scope of this article; *not* building information modelling)
S2-C	Clinical Outcome Monitor, a clinical-phase agent (outside the scope of this article)
Artificial intelligence and information retrieval	
AI	Artificial intelligence
LLM	Large language model
RAG	Retrieval-Augmented Generation
GPT	Generative pre-trained transformer (family of the cloud reference models used)

Table S14 (continued)

Abbreviation	Expansion and meaning in this article
ASA	Agent-Based Service Architecture, the closest published precedent used as comparison
TARAG	Time-Aware Agricultural RAG, a compared framework from the literature
Hit@k	Fraction of queries whose ground-truth document appears within the top k retrieved
MRR	Mean reciprocal rank of the first relevant retrieved document
nDCG@K	Normalized discounted cumulative gain over the top K retrieved documents
F1	Harmonic mean of precision and recall
MAE	Mean absolute error
RMSE	Root-mean-square error
CI	Confidence interval (Wilson score interval for binomial proportions)
κ	Cohen's kappa, chance-corrected inter-rater agreement
Regulatory quality frameworks	
GACP	Good Agricultural and Collection Practice (medicinal-plant starting materials)
GMP	Good Manufacturing Practice
GLP	Good Laboratory Practice
GCP	Good Clinical Practice
SOP	Standard operating procedure
QA	Quality assurance
EMA	European Medicines Agency
HMPC	Committee on Herbal Medicinal Products (of the EMA)
RCT	Randomized controlled trial
DSMB	Data Safety Monitoring Board
Sensing, edge computing and data	
IoT	Internet of Things

Table S14 (continued)

Abbreviation	Expansion and meaning in this article
MQTT	Message Queuing Telemetry Transport, the publish–subscribe protocol used for telemetry
ESP32	Microcontroller family used in the sensing nodes
API	Application programming interface
JSON	JavaScript Object Notation, the serialization format of the traceability records
QR	Quick-response two-dimensional barcode, used to link physical batches to digital records
VPD	Vapour-pressure deficit
DLI	Daily light integral
EC	Electrical conductivity (of the fertigation solution)
LED	Light-emitting diode
GPU	Graphics processing unit
SHA-256	Secure Hash Algorithm, 256-bit variant
keccak256	Keccak hash function, 256-bit output, as used by EVM-compatible networks
EVM	Ethereum Virtual Machine, the execution environment of the compatible notarization networks
Chemistry, institutions and funding	
THC	Δ^9-tetrahydrocannabinol
CBD	Cannabidiol
CONICET	National Scientific and Technical Research Council (Argentina)
CENPAT	National Patagonian Center (CONICET)
TYS	Technology and Health research group, University of Alicante
CRediT	Contributor Roles Taxonomy, used for the authorship statement